Hand-Crafted Shape and Texture Descriptors for Cats vs. Dogs: Roberts-Cross HoG + Global LBP with an RBF SVM

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Abstract

Before deep learning, image classification relied on hand-designed descriptors of shape and texture. In this project we build such a pipeline from scratch and measure how far it goes on a task that is easy for people: telling cats from dogs.

*We implement a modified Histogram of Oriented Gradients (HoG) that computes image derivatives with the 2×2 Roberts-cross operator instead of the usual central differences, and a global Local Binary Pattern (LBP) descriptor made of regional LBP histograms. Concatenated, they form a 5,860-dimensional feature vector that captures contour shape and fur texture. An RBF-kernel SVM trained on a balanced 10,000-image subset reaches **78.05% test accuracy** with almost identical error rates for the two classes. We analyse what each descriptor contributes and where hand-crafted features reach their limits.*

1. Introduction

Why hand-crafted features? Descriptors such as HoG [3] and LBP [6] are fast, need no GPU, and are fully interpretable: every dimension has a geometric meaning. Building them from first principles shows what information a classifier actually receives, and gives a baseline against which learned features can be judged.

Our approach. We combine two complementary cues:

- **Shape**, via a HoG variant with Roberts-cross derivatives [7], which respond to the two diagonal directions with a very compact 2×2 support;
- **Texture**, via LBP codes pooled into a 4×4 grid of histograms, following the regional LBP descriptor of Ahonen *et al.* [1].

The concatenated vector is classified by a support-vector machine [2].

2. Related Work

HoG. Dalal and Triggs [3] showed that histograms of gradient orientation, normalised over overlapping blocks, describe object shape robustly; their study found that simple centred derivatives work better than larger smoothing masks.

LBP. Ojala *et al.* [6] encode each pixel by thresholding its neighbours against it; histograms of these codes are strong grayscale texture descriptors. Ahonen *et al.* [1] concatenated regional histograms to keep coarse spatial layout.

Cats vs. dogs. The task comes from the Asirra CAPTCHA [4], which was designed to be hard for machines. Golle [5] reported about 83% accuracy with SVMs on colour and texture features, which is a useful reference point for hand-crafted pipelines.

3. Method

3.1. Modified HoG with Roberts-cross derivatives

Images are resized to 64×64 and converted to grayscale. The derivatives are

$$K_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad K_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \tag{1}$$

$$|G| = \sqrt{G_x^2 + G_y^2}, \qquad \theta = \operatorname{atan2}(G_y, G_x) \bmod 180^\circ. \tag{2}$$

Each 8×8-pixel cell accumulates a 9-bin histogram of unsigned orientation (20° bins) weighted by $|G|$. Blocks of 2×2 cells with a one-cell stride (7×7 blocks) are L2-Hys normalised: L2-normalised, clipped at 0.2 and normalised again. This gives $7 \cdot 7 \cdot 36 =$ **1,764** dimensions.

Because the Roberts kernels measure differences along the diagonals, θ is offset by 45° from the image axes. For an orientation histogram this only shifts the bin boundaries, so it does not reduce the descriptor's power.

3.2. Global LBP descriptor

For each pixel with centre value g_c and its $P = 8$ neighbours g_p at radius $R = 1$,

$$\mathrm{LBP}_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c)\, 2^p, \qquad s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases} \tag{3}$$

The LBP image is split into a 4×4 grid. Each region contributes a normalised 256-bin histogram, and the 16 histograms are concatenated into **4,096** dimensions that describe texture while keeping coarse layout.

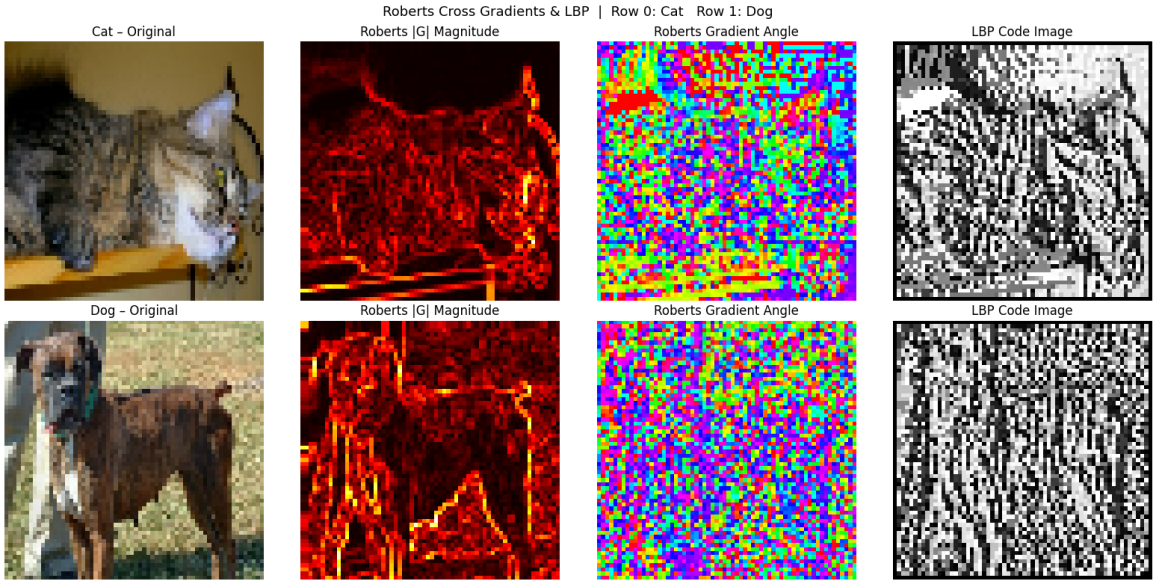


Figure 1. For one cat (top) and one dog (bottom): input, Roberts gradient magnitude, gradient orientation and LBP code image. Magnitude highlights contours and facial features; LBP highlights fur texture.

Table 1. Test-set classification report (2,000 images).

Class	Precision	Recall	F1	Support
Cat	0.78	0.78	0.78	1,000
Dog	0.78	0.78	0.78	1,000
Accuracy			**0.7805**	2,000

3.3. Classifier

The two descriptors are concatenated (5,860 dimensions), standardised with statistics from the training split only, and classified by an SVM with an RBF kernel ($C = 10$, $\gamma = \texttt{scale}$).

4. Results and Discussion

4.1. The data

We use the Cats vs. Dogs dataset [4] (via TensorFlow Datasets). Extracting features from all 25k images exceeded the available memory, so we draw a balanced random subset of 5,000 cats and 5,000 dogs and split it 80/20 (stratified, seed 42) into 8,000 training and 2,000 test images. In the released code each of the 256 LBP codes has its own histogram bin; the run reported here used a bin range that merged codes 254 and 255, a negligible difference.

4.2. Quantitative results

The SVM reaches **78.05%** test accuracy (Table 1). The confusion matrix (Fig. 2) shows 778 of 1,000 cats and 783 of 1,000 dogs classified correctly.

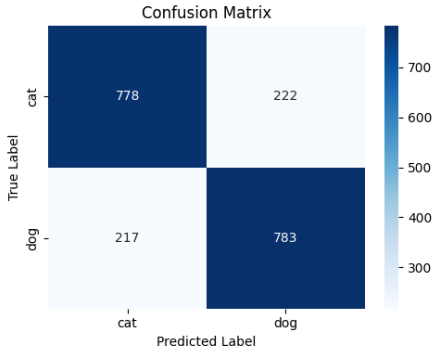


Figure 2. Confusion matrix on the test set.

4.3. Analysis

Balanced errors. The two error types are almost equal (222 cats predicted dog, 217 dogs predicted cat), so the classifier is not biased toward either class.

Complementary cues. The Roberts magnitude (Fig. 1) traces ears, eyes, snout and body outline, which is the information HoG summarises. The LBP image responds to fur and background texture, which HoG ignores. An ablation (HoG only, LBP only, both) was not run; it would quantify each descriptor's contribution.

Where the pipeline loses. The 2×2 Roberts kernel is more sensitive to noise than a 3×3 Sobel operator, which plausibly hurts on cluttered backgrounds. Resizing to 64×64 discards fine detail. Neither descriptor uses colour, which Golle [5] found helpful. Fine-tuned CNNs exceed 95% on this task, which marks the gap between hand-crafted and learned features.

5. Conclusion & Future Work

A from-scratch Roberts-cross HoG combined with a regional LBP descriptor and an RBF SVM classifies cats and dogs with 78.05% accuracy and balanced errors. That is reasonable for purely hand-crafted, grayscale features.

Next steps. (i) Run the HoG/LBP ablation; (ii) apply a light Gaussian blur before differentiation to reduce Roberts-kernel noise; (iii) use 128×128 inputs together with PCA to control dimensionality; (iv) add colour histograms; and (v) use the full 25k images with a linear SVM or incremental training to avoid memory limits.

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